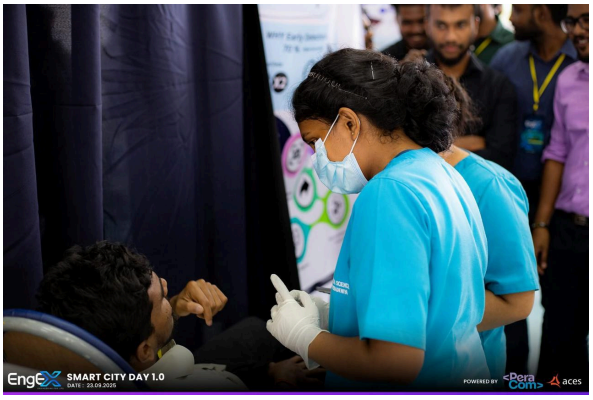
Volunteer Contributions

OASIS Dental Application & AI Oral Cancer Classification

- Developed the Flutter frontend for OASIS Dental Application and integrated it with a Spring Boot backend.
- Hosted the backend on the department server and deployed the system for real-time dental checkups during Engex 2025.
- Developed a backend server integrating an AI model created by E17 batch to classify oral cancer using metadata and mouth images.
- Hosted the AI classification backend on the Tesla server.
- Presented the system at Engex Exhibition 2025.
- Led the organization and arrangement of the dental stall where live oral checkups were conducted with participation from Dental Faculty students and a doctor to provide live feedback through the app thereby emphasizing the practical aspect.

GitHub:

- https://github.com/Oral-Cavity-Research/OASIS-flutter-frontend.git
- https://github.com/fikrynuhuman183/OASIS-my-dev.git



Engex 2025 Crowd Management System

- Integrated and hosted the Engex 2025 crowd management application under multiple operational constraints.
- Managed GitHub repository with multiple versions and deployments.
- Hosted system using a dedicated PC as server.
- Configured NAT, SSL (via NCSU), and port forwarding.
- Deployed multi-backend + frontend integrated system with kiosk support.

Active Research Involvement

ESCAL (Since 2025)

- Involved in **Peraswarm research**.
- Actively participating in **Neuromorphic research**.

MARC – Lung Disease Research

- Active undergraduate researcher contributing to lung disease research initiatives.

Casual Instructor – Faculty of Engineering

Courses:

- CO1010 – Computer Programming
- CO1020 – Computer Systems Programming
- CO2040 – Software Design and Development (Present)
- CO2060 – Software Architecture (2nd Year Project) (Present)

Career Fair – E20 (Trace City)

- Assisted in coordination of the Career Fair.
- Lead Coordinator for GTN.

Project Nenathambara (Since 2025)

- Active participant.
- Conducted training programs for school teachers on Arduino and programming.
- Managed organizing and distribution of Arduino kits to multiple schools.

E22 Welcome Ceremony

- Designed and printed file covers and banners.

Ruusara 2024 – Faculty of Engineering Website

- Contributed to website design and development.
- Repository: https://github.com/ruusara24/ruusara2024